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SUMMARY

Computer Science Engineering student building developer tooling, simulations and Android apps, and fixing bugs upstream in AI tooling: 14 pull requests merged into projects I don't own, including Microsoft PyRIT and UK AISI Inspect AI. I like finding out how systems behave beneath the surface.

EDUCATION

B.Tech in Computer Science and Engineering *SCMS School of Engineering and Technology*
CGPA: 8.11/10 (after Semester 2)

TECHNICAL SKILLS

Languages	Python, Java, Kotlin, TypeScript, JavaScript, SQL, HTML/CSS
Frameworks & libraries	three.js (WebGL, GLSL), Jetpack Compose, Java Swing with JavaFX WebView, Apache PDFBox, Vite
Databases	MySQL (JDBC, transactions, row locking), SQLite
Testing	JUnit 5, pytest, Hypothesis, Vitest, Playwright
Tools & infrastructure	Git, GitHub Actions, Gradle, Maven, Vercel, GitHub Pages, Cloudflare Workers

SELECTED PROJECTS

Event Horizon *TypeScript · three.js · GLSL · Vite · Vitest* — live demo · source

- Built a fixed-timestep simulation core (semi-implicit Euler with Plummer softening, direct-sum body gravity, struct-of-arrays particle field) on a seeded RNG, so the same seed reproduces the same universe.
- Wrote 105 passing unit tests, including momentum conservation through black-hole mergers (under 1e-6 relative drift over 3,600 steps) and same-seed determinism; wrote custom GLSL shaders for the photon ring and limb darkening.

Campus Bus *Java 17 · Swing + JavaFX WebView · MySQL 8 · JUnit 5* — Windows release v1.1 · source

- Kept seat booking correct under concurrent requests with transactional `SELECT … FOR UPDATE` row locks, covered by seat-race tests; generated PDF passes with PDFBox and emailed them over SMTP.
- Wrote 166 tests (87 run without a database, the rest against an isolated MySQL 8.4), with one regression test per pre-release audit defect; packaged a self-contained Windows build with `jlink` and a SHA-256 checksum.

Pipo *Kotlin · Jetpack Compose · Cloudflare Workers · three.js* — APK release v1.0 · source

- Built a deterministic life simulation (behaviour, mood, world and trip engines) that advances while the app is closed and reports what happened on return; 226 JVM unit tests, including seeded long-run simulations.
- Kept LLM provider keys out of the APK by routing optional Gemini/Groq chat through a Cloudflare Worker proxy, with an offline rule-based brain as the default.

gh-ops *Python · GitHub Actions · pytest · Hypothesis* — source

- Runs six scheduled jobs (CI, release, endpoint and security monitoring, plus daily and weekly reports) through one dispatcher on Actions cron, with read-only collectors, state snapshots and change detection; pings Telegram only when something changed. 1,499 tests passing in CI.

OPEN-SOURCE CONTRIBUTIONS

Microsoft PyRIT *AI red-teaming framework* — 5 merged PRs

- Added read-only dataset summaries (counts, modalities, harm categories) computed from stored seeds without loading values or calling providers, for a planned dataset explorer; 11 files, with tests (#2762).
- Fixed four correctness bugs, each with tests: XXE scorer missing XInclude attacks (#3429), exact-text matching treating an empty target as a match (#2882), the HarmBench loader dropping context strings (#2823), and converter identifiers colliding across word-selection strategies (#2976).

Inspect AI *UK AI Security Institute, LLM evaluation framework* — 1 merged PR

- Made an explicitly passed Gemini API key take precedence over ambient `GOOGLE_USE_ADC` credentials, while keeping explicit `use_adc=True` as an override; fixes #5358, with tests (#5670).

aios *open-source agent runtime* — 4 merged PRs

- Preserved `finish_reason="length"` when streaming trailer chunks overwrote it, and recorded truncation on the model-request span (#2457, #2458); made workflow timeouts report whether the deadline or the spend bound fired (#2459); stopped the harness bypassing LiteLLM parameter translation (#2460).

14 pull requests merged into projects I don't own. Full ledger: `github.com/aspire488/oss-atlas`